

DIYA KINI

Austin, TX | (512) 944-3923 | [Email](#) | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

University of Michigan

Ann Arbor, MI

B.S.E in Computer Science, Robotics

2023 - 2026

- Academic Honors: Michigander Scholar, Dean's Honor List, Amazon's Women of the World, Live Más Scholar
- Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Computer Vision, Discrete Mathematics

EXPERIENCE

Lockheed Martin (LAIC)

Remote

Artificial Intelligence Intern

May 2025 - Present

- Applied spatiotemporal deep learning models (STGCN, GAT, TGN) in Python to analyze and cluster aircraft trajectory data, conducting experiments that identified the most accurate model for predictive modeling
- Containerized the ML workflow leveraging Podman, Poetry, and DevContainers to ensure reproducible deployment
- Developed a Python WebSocket communication layer to interface with a C++ Open Mission Systems (OMS) server, enabling CAL compatibility and facilitating seamless integration with next-generation avionics platforms

University of Michigan Robotics – The MBot Ecosystem

Ann Arbor, MI

Research Assistant

Jan 2024 - May 2024

- Led autonomous robotic system development and improvement for 100+ vision-controlled robots
- Applied mathematical operations to Python, OpenCV, and computer vision algorithms to facilitate distance learning
- Enhanced the Jetson Nano's stereo vision implementation within the MBot ecosystem through ORB SLAM
- Integrated the updated vision systems in 2 University of Michigan's robotics courses, impacting 40 graduate students

University of Michigan College of Engineering

Ann Arbor, MI

Computational Linear Algebra Instructional Aide

Jan 2024 - Present

- Executed matrix operations and linear transformations to optimize LiDAR data interpretations in robotic navigation
- Taught 200+ students the application of ML concepts, including linear regression and classification, to develop predictive models for robotic systems, enhancing pattern recognition and decision-making processes
- Explained feedback control algorithms for dynamical systems, leveraging linear algebra and Julia programming

Berry Consultants, LLC

Austin, TX

Data Science Intern

June 2022 - Aug 2022

- Analyzed the optimal length of time a novel gene therapy would keep hemophilia patients at a safe level of clotting
- Assisted in the FDA approval of 2 treatments by investigating the relationship between biomarkers of the gut biome and clinical outcomes for an innovative treatment for C-difficile infections through the R programming language

PROJECTS

Medical Liquid Dispenser (CV & Robotics)

Feb 2025

- Integrated computer vision into an automated liquid dispenser, achieving 95% accuracy in hand gesture recognition using MediaPipe Hands for real-time control of dispensing, allowing users to select liquid portions with hand signals
- Improved user interaction by enabling hands-free operation with 86% reliability in various lighting conditions

Predicting FX Rates with Sentiment Analysis (TensorFlow)

July 2024

- Used time-series analysis and LSTMs to predict EUR/USD exchange rate from existing data in Yahoo Finance
- Incorporated news sentiments from past news articles to improve forecast accuracy through NLP techniques and Tensorflow by analyzing the emotional tone of words in the text discussing historical conflicts between nations

ML Robot Tour (Python)

Dec 2023

- Integrated vision-based navigation system, enabling adaptive robot pathfinding upon numerical cue detection
- Optimized route selection of MBot Omni's obstacle avoidance by an increase of 22% through SLAM and BFS
- Trained 3 ML numerical recognition models (KNN, Linear Classifier, NN) using SciKit Learn and MNIST database

SKILLS

Licenses & Certifications:

TensorFlow Professional Developer Certificate (DeepLearning.AI),
Quantitative Modeling for Analysts Specialization (Wharton Online)

Programming Languages & Computer Skills:

Python, C++, R, Java, Solidworks, Julia, OpenCV, ROS, MATLAB, Verilog